



SRF735 THRU SRF760(SINGLE CHIP)

7.5AMPS. SCHOTTKY BARRIER RECTIFIERS

FEATURES

• Plastic package has Underwriters Laboratory Flammability Classification 94V – O

• Metal silicon junction, majority carrier conduction

• Guard ring for overvoltage protection

• Low power loss, high efficiency

• High current capability, Low forward voltage drop

• Single rectifier construction

• High surge capability

• For use in low voltage, high frequency inverters, free wheeling, and polarity protection applications

• High temperature soldering guaranteed: 250°C/10 seconds,

• 0.25"(6.35mm) from case

MECHANICAL DATA

• Case: JEDEC ITO – 220AC molded plastic body

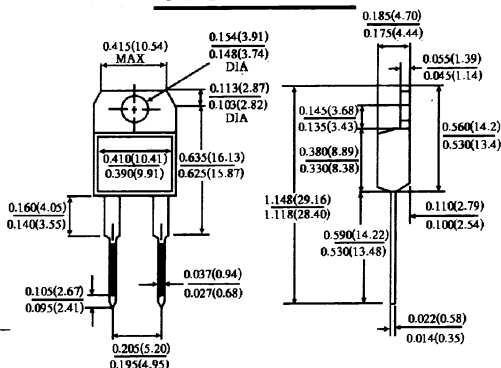
• Terminals: Lead solderable per MIL – STD – 750, method 2026

• Polarity: As marked

• Mounting Position: Any

• Weight: 0.08ounce, 2.24 gram

1TO-220AC



MAXIMUM RATINGS AND ELECTRICAL CHARACTERISTICS

(Ratings at 25°C ambient temperature unless otherwise specified, Single phase, half wave, resistive or inductive load. For capacitive load, derate by 20% .)

		Symbols	SRF735	SRF745	SRF750	SRF760	Units
Maximum repetitive peak reverse voltage		V _{RRM}	35	45	50	60	Volts
Maximum RMS voltage		V _{RMS}	25	32	35	42	Volts
Maximum DC blocking voltage		V _{DC}	35	45	50	60	Volts
Maximum average forward rectified current (see Fig.1)		I _(AV)	7.5				Amps
Repetitive peak forward current(square wavr, 20KHZ) at Tc = 105°C		I _{FRM}	15.0				Amps
Peak forward surge current 8.3ms single half sine – wave superimposed on rated load (JEDEC method)		I _{FSM}	150.0				Amps
Maximum instantaneous forward voltage at 7.5 A(Note 1)		V _F	0.65		0.75		Volts
Maxlimum instantaneous reverse current at rated DC blocking vol-fage(Note 1)	T _A = 25°C	I _R	1.0			mA	
	T _A = 125°C		15		50		
Typical thermal resistance (Note 2)		R _{θJC}	5.0				°c/w
Operating junction temperature range		T _J	– 65 to + 150				°C
Storage temperature range		T _{STG}	– 65 to + 150				°C

Notes: 1. Pulse fest: 300 μ s pulse width, 1% duty cycle

2. Thermal resistance from junction to case

RATINGS AND CHARACTERISTIC CURVES(SRF735 THRU SRF760)

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FIG. 1 – FORWARD CURRENT DERATING CURVE

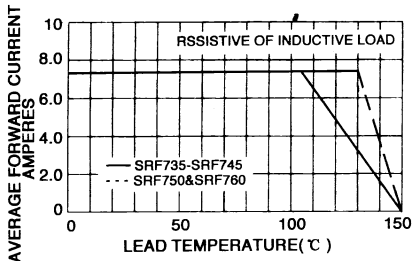


FIG. 2 – MAXIMUM NON - REPETITIVE PEAK FORWARD SURGE CURRENT

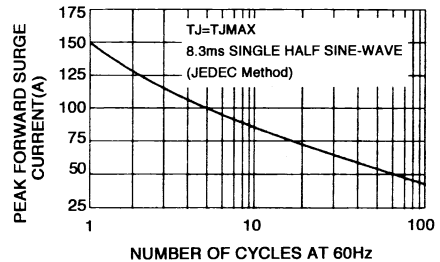


FIG. 3 – TYPICAL INSTANTANEOUS FORWARD CHARACTERISTICS

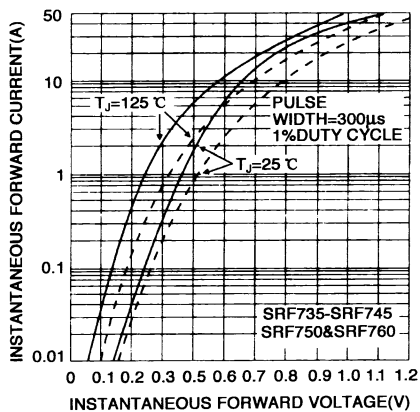


FIG. 4 – TYPICAL REVERSE CHARACTERISTICS

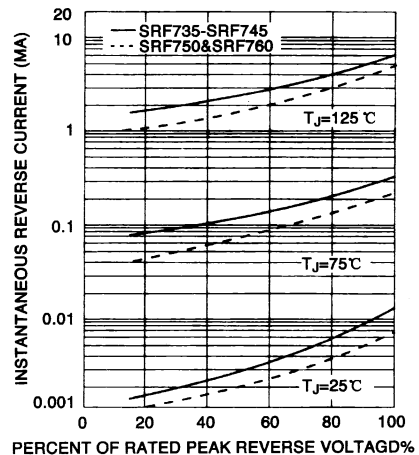


FIG. 5 – TYPICAL JUNCTION CAPACITANCE

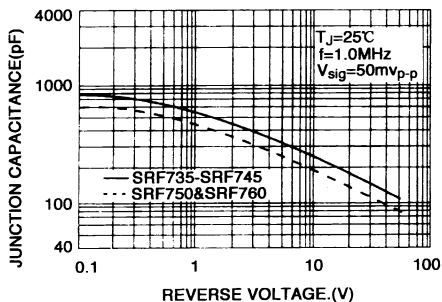


FIG. 6 – TYPICAL TRANSIENT THERMAL IMPEDANCE

